
Feedback or Annealing?

A Controlled Study of Adaptive State Aggregation

Sunny Zhang
University of Toronto
ssunny.zhang@mail.utoronto.ca

Abstract

Adaptive state aggregation replaces full-state Bellman updates with updates on groups of states having similar values. A natural refinement rule shrinks the group width in response to the Bellman residual, but any gain may come from annealing itself rather than from feedback. I separate the two on a 9,261-state inventory-control MDP. Because the Bellman operator contracts the residual span by γ per sweep, the residual span, and therefore the residual-driven aggregation also decays geometrically. This gives a natural open-loop baseline whose decay rate is determined in advance by γ and the solver's update schedule, rather than tuned after observing the adaptive method. Across three global/aggregate update mixes, 20 paired sampling seeds, and five compute budgets, feedback's largest reliable advantage over that timetable is 0.3% of the control's error and its largest disadvantage is 14.5%; every 95% bootstrap interval for the final paired difference lies inside a preregistered ± 0.02 practical-equivalence region. The schedule feedback induces is worth far more than the feedback: against a fixed $\varepsilon = 0.05$ it moves the error by up to 23.5% at a matched budget, in a direction that reverses as the budget grows, so the final iterate alone reports neither effect. Full-state value iteration also reaches the ground-truth tolerance in fewer charged backups than any aggregation arm. The case study shows why adaptive dynamic programming needs fixed-resolution, matched open-loop, and decision-quality controls, at more than one budget.

1 Introduction

State aggregation combats the state-space cost of dynamic programming by solving a smaller surrogate MDP [1–3]. In the adaptive scheme of Chen et al. [4], states with nearby current values share a group, and ordinary Bellman sweeps alternate with stochastic group sweeps. A fixed bin width ε creates a familiar cost–accuracy tradeoff: coarse groups are cheap but impose approximation error, while fine groups preserve accuracy but perform more backups.

This invites a feedback controller. When the Bellman residual is large, a coarse representation may suffice; as the residual contracts, the algorithm can refine. The key identification problem is that feedback also creates an annealing schedule. Comparing it only with a fixed width cannot tell whether the residual signal helps or whether almost any schedule with the same decay would work.

I test the signal itself, against an open-loop geometric control matched in starting width, floor, and approximate time to the floor. The original experiment was preregistered at one update mix and a 20 ms endpoint; I preserve that result, including the endpoint's failure, and add a work-indexed robustness check at three mixes. Three things come out of it. First, the matched control is a consequence, not a modelling choice: span contraction makes the residual-driven width obey a geometric envelope with ratio γ per global sweep (Proposition 1), so feedback cannot produce a schedule shape an open-loop rule could not. Second, the signal buys nothing at any of five compute budgets; where the arms separate, feedback is worse. Third, the schedule it induces is not innocuous:

at a matched budget it moves the error by up to 23.5% relative to never refining, in a direction that reverses as the budget grows. A final-iterate comparison reports none of this.

2 Related work

Classical hard aggregation replaces each block of states by a common value, and approximation and induced-policy losses relate to within-block value variation [1, 5]; abstraction theory separates value-, action-, reward-, and transition-based notions of similarity [3]. Several methods adapt the representation during computation: Bertsekas and Castanon [2] group states by residual magnitude, Singh et al. [6] adapt soft cluster memberships, variable-resolution control splits cells on value, policy, influence, or variance criteria [7], and progressive disaggregation slices regions relative to the evolving value function [8]. I use the simpler construction of Chen et al. [4], which repeatedly bins states by current value and guarantees asymptotic error $2\varepsilon/(1 - \gamma)$ when within-bin variation is at most ε .

Bellman residuals provide computable value and greedy-policy error certificates [9]; their span seminorm has been optimized directly in approximate value iteration [10], and prioritized sweeping uses local errors to choose backups [11, 12]. Yet a small residual need not be a reliable proxy for the decision objective under approximation [13], motivating my separate policy-loss evaluation. My use of the residual is narrower: its global span changes one resolution parameter, not group membership, state priority, or the fitting objective. I propose no new abstraction or convergence theorem; I test whether that feedback improves the finite-compute trajectory once the open-loop annealing path it induces is controlled for.

3 Method

For a finite discounted cost MDP, define

$$(TV)(s) = \min_a \left[c(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right]. \quad (1)$$

The solver bins $V(s)$ into occupied intervals of width ε . A global iteration evaluates (1) for every state. An aggregate iteration samples one state per group, performs one lifted Bellman backup for that representative, and updates the group value with step size $\alpha_t = 1/\sqrt{t}$. A schedule (ℓ_g, ℓ_a) repeats ℓ_g global and ℓ_a aggregate iterations. Rebinning occurs on entry to each aggregate phase, and lifting occurs on return to the global phase.

After each global sweep I record $r_t = \text{span}(TV_t - V_t) = \max_s (TV_t - V_t)(s) - \min_s (TV_t - V_t)(s)$. The tested feedback rule is

$$\varepsilon_t^{\text{RESIDUAL}} = \max\{\varepsilon_{\min}, cr_t\}, \quad c = 0.084, \quad \varepsilon_{\min} = 0.05. \quad (2)$$

The constant aligns the first residual-controlled width with the coarsest fixed baseline: after the two arm-independent opening sweeps, $r_0 = 5.93429$ deterministically, so $0.084r_0 = 0.49848 \approx 0.5$. The rule reaches the floor after roughly 14 aggregate entries. The main control never reads r_t :

$$\varepsilon_j^{\text{GEOMETRIC}} = \max\left\{\varepsilon_{\min}, 0.4985 \left(\frac{\varepsilon_{\min}}{0.4985}\right)^{j/13}\right\}. \quad (3)$$

Thus RESIDUAL and fast GEOMETRIC differ primarily in closed-loop versus open-loop refinement. A fixed $\varepsilon = 0.05$ arm tests whether refinement is useful at all. Slower geometric and coarser fixed controls are retained in the artifact but are not needed for the central contrast.

Why a geometric control is the right one. Matching GEOMETRIC to RESIDUAL is not a curve fit. The shape is forced.

Proposition 1. *Along the global sweeps of any schedule, $r_{t+1} \leq \gamma r_t$. Hence $\varepsilon^{\text{RESIDUAL}} \leq \max\{\varepsilon_{\min}, \varepsilon_0^{\text{RESIDUAL}} \gamma^{n_g}\}$, where n_g counts global sweeps since the first aggregate entry, and the width reaches its floor within $\lceil \log(\varepsilon_{\min}/\varepsilon_0^{\text{RESIDUAL}}) / \log \gamma \rceil$ of them.*

Table 1: Final iterate over 20 paired seeds. Columns R/G/F are residual, fast geometric, and fixed $\varepsilon = .05$. The interval is for the median paired error difference R–G. Policy loss and billed backups are reported in the same order; B(M) denotes millions of billed backups.

Mix	$\text{err}_\infty(\text{R})$	$\text{err}_\infty(\text{G})$	$\text{err}_\infty(\text{F})$	R–G (95% CI)	Policy loss R/G/F	B(M) R/G/F
(5, 2)	.05001	.04368	.05001	.00632 [.00632,.00633]	.00210/.00278/.00210	67.685/67.694/67.686
(2, 5)	.13882	.13882	.13791	–.000003 [–.000412,.000000]	.01518/.01609/.01280	30.214/30.212/30.195
(1, 20)	.20343	.20053	.19093	–.00018 [–.00208,.00609]	.05178/.05178/.01649	9.185/9.193/9.196

Note: R–G is the median of the 20 seedwise paired differences, not the difference of the marginal medians, so at (1, 20) it can be negative while R’s displayed median exceeds G’s.

The proof (Appendix A) is span contraction of T ; the aggregate phase is not covered, so the claim is an envelope over the stretch in which the rule is still setting the width. That is the stretch that matters, and it holds at every recorded iterate of every mix: the realized per-cycle decay of $\varepsilon^{\text{RESIDUAL}}$ is 0.698, 0.829, and 0.829 at (5, 2), (2, 5), and (1, 20), against bounds γ^{ℓ_g} of 0.774, 0.903, and 0.950, while GEOMETRIC’s designed ratio of 0.838 per aggregate entry sits within 1% of the realized figure at two of the three mixes. A closed-loop rule reading a contracting residual is therefore an annealing schedule whose rate is pinned by γ and ℓ_g before the first run, which is what makes this an identification problem rather than a horse race. The same fact bounds the matching: the first aggregate entry follows ℓ_g global sweeps, so RESIDUAL starts at $\varepsilon_0^{\text{RESIDUAL}} = 0.400, 0.499$, and 0.538 across the mixes while GEOMETRIC is anchored at 0.4985 throughout. The arms share an initial width only at the preregistered (2, 5); at (5, 2) GEOMETRIC begins 25% coarser, which plausibly accounts for part of the gap reported below.

Inventory MDP. A state is the inventory vector $q \in \{-10, \dots, 10\}^3$, giving $|\mathcal{S}| = 21^3 = 9,261$. Five joint quote-aggressiveness actions trade fill probability against spread; a transition is no fill or a one-unit fill on either side of one asset, and the one-period cost combines spread revenue with correlated inventory risk (Appendix C). I use $\gamma = .95$ and rescale costs so $\|V^*\|_\infty = 100$. This is a small market-making-style control model, not a calibrated market simulator.

Protocol and metrics. Each arm runs 10,000 iterations at (5, 2), (2, 5), and (1, 20). The MDP is deterministic; seeds 0–19 vary the within-group sampling stream and are paired across arms. The primary metric is $\text{err}_\infty = \|V - V^*\|_\infty$; I also evaluate the greedy policy π_V by $\|V^{\pi_V} - V^*\|_\infty$. Work-indexed curves use billed Bellman backups $B = n_g|\mathcal{S}| + \sum_t K_t$ for K_t occupied groups, an accounting favorable to aggregation: rebinning and lifting add $(n_r + n_l)|\mathcal{S}|$ state operations, and the residual scan is not charged. Paired differences are reported as a median with a seeded 10,000-resample percentile-bootstrap interval, against the preregistered practical-equivalence region $[-0.02, 0.02]$ on the 100-normalized value scale.

4 Results

Once the residual and geometric rules reach $\varepsilon_{\min}$, their work–error trajectories are visually inseparable at every update mix (Figure 1, Appendix B). Table 1 makes the small differences visible. At all three mixes, the full paired interval for residual minus fast geometric lies inside the ± 0.02 equivalence region. At (5, 2) the interval excludes zero and every seed favors geometric in value error, but the median difference is only 0.00632. At the preregistered (2, 5) mix the median difference is -2.9×10^{-6} , with an interval straddling zero. Residual feedback is therefore never practically better and is sometimes detectably, though not practically, worse.

The fixed control adds a qualification: value accuracy does not order decision quality. At (5, 2) fast geometric has the lowest value error but higher policy loss than residual, and at (2, 5) the slow geometric arm (artifact only) has the lowest value error and 2.5 times fixed-0.05’s policy loss. At (1, 20) residual trails fixed-0.05 on both (+0.01250 in value error, 95% CI [0.00745, 0.01561]; +0.03529 in policy loss). Feedback cannot be justified by either metric here.

The mixes also consume different work. Within a mix the arms’ median final work differs by at most 0.13%, so the final-iterate comparisons are effectively work-matched, but representation maintenance adds 39%, 88%, and 96% for the residual arm across the three mixes. The global-heavy mix’s lower final error is therefore not a “schedule wins” result: it purchases more global work. Full-state VI is

the sharper baseline: the same implementation meets its 10^{-10} tolerance in 454 sweeps, or 4.20M billed backups. On this instance aggregation is not competitive; the experiment identifies behavior inside an approximation scheme, not a new best solver.

Which decision was the schedule, and which was the feedback? The final iterate says whether feedback moves the endpoint, not whether refinement scheduling does anything at all: by 10,000 iterations every scheduled arm has sat at $\varepsilon_{\min}$ for almost the whole run, and at (5, 2) the residual and fixed arms end 3.8×10^{-8} apart, a comparison with nothing left to compare. Re-reading the same runs at four intermediate budgets, where the schedules are still live, separates the two decisions (Appendix D; differences are relative to the control, since the $\pm .02$ margin says nothing at a budget where both arms sit at error 10). Feedback’s largest reliable advantage over the matched timetable at any budget is 0.3% of the control’s error, its largest disadvantage 14.5%. Refinement scheduling is worth up to 23.5%, two orders of magnitude more, and it reverses: at (1, 20) annealing is 18.8% better than never refining at 10^5 backups and 10.2% worse at 3×10^6 . The schedule does real work in both directions while the feedback generating it contributes nothing, exactly as Proposition 1 predicts. The large RESIDUAL–FIXED entries fall in the steep descent, where a fixed budget lands between iterates; their sign and order of magnitude are robust, their exact percentages are not.

The preregistered endpoint failed. The original primary endpoint was (2, 5) at 20 ms. It gave a residual-minus-geometric median of -0.0417 with 95% CI $[-0.0459, 0.0011]$, indeterminate under the equivalence rule. Worse, repeated executions changed the interval because 20 ms lies on a steep transient and selects a different iterate as machine load changes. I report rather than replace that outcome. Every analysis indexed by backups is explicitly a robustness analysis motivated by this failure, not a retroactively preregistered endpoint.

5 What the negative result isolates

Endpoint matching is load-bearing. Transplanted unchanged to (1, 20), the slow geometric arm never finishes its timetable and ends ten times worse than every other arm; recalibrating only its cycle count restores it (Appendix E). An open-loop control must be matched in the clock on which adaptation occurs: here aggregate entries, not total iterations.

Equivalence is not equality. At (5, 2) the paired interval $[.006319, .006327]$ is narrow enough to exclude zero, so calling the arms “indistinguishable” would be false; the predeclared question was whether the whole interval lies inside a practically negligible $\pm .02$ region, and it does. The (1, 20) interval instead crosses zero but is wider. Both satisfy the same practical criterion for opposite statistical reasons, which is the trap in negative results with low simulation variance: abundant precision detects differences far too small to justify the mechanism.

Artifact and audit trail. The supplement carries the preregistration frozen before the residual arm ran, the sweeps, per-seed endpoints, realized width paths, environment, and a test suite covering the $\ell_g = 1, \ell_a = 0$ reduction to ordinary VI.

6 Limitations and conclusion

This study has one deterministic inventory instance, one residual statistic, one floor, and one feedback constant, so no claim is made about other MDPs or about settings where aggregation genuinely beats VI; the mixes, backup endpoints, and budget ladder were added after the preregistered timing experiment. Billed backups omit residual computation and representation maintenance, which only strengthens VI but may reorder the aggregation arms. Proposition 1 marks its own boundary: feedback has room to matter where the envelope stops being tight: heterogeneous regional residuals, an unknown floor, or a horizon ending before the schedule does.

Within this scope the result decomposes. Residual feedback never practically improves on a matched open-loop timetable, and Proposition 1 says why: a rule reading a contracting residual is already an annealing schedule, one worth up to 23.5% at a matched budget, with a budget-dependent sign. Attributing that gain needs controls reproducing the path the feedback induces, and more than one endpoint; otherwise ordinary scheduling is mistaken for intelligence.

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A Proof of Proposition 1

Fix V, U and a state s , and let a^* attain the minimum in $(TU)(s)$. Evaluating the bracket of (1) at a^* rather than at the minimizer for V gives

$$\begin{aligned} (TV)(s) &\leq c(s, a^*) + \gamma \sum_{s'} P(s' | s, a^*) V(s') = (TU)(s) + \gamma \sum_{s'} P(s' | s, a^*) (V - U)(s') \\ &\leq (TU)(s) + \gamma \max_{s'} (V - U)(s'), \end{aligned}$$

and the symmetric argument, taking a^* to attain the minimum in $(TV)(s)$, gives $(TV)(s) \geq (TU)(s) + \gamma \min_{s'} (V - U)(s')$. Both bounds hold at every s , so $(TV - TU)(s)$ lies in $[\gamma \min(V - U), \gamma \max(V - U)]$ and therefore

$$\text{span}(TV - TU) \leq \gamma \text{span}(V - U).$$

Along a global phase $V_{t+1} = TV_t$, so $r_{t+1} = \text{span}(TV_{t+1} - V_{t+1}) = \text{span}(TV_{t+1} - TV_t) \leq \gamma \text{span}(V_{t+1} - V_t) = \gamma \text{span}(TV_t - V_t) = \gamma r_t$. Iterating over the n_g global sweeps taken since the first aggregate entry and substituting into (2) gives the stated envelope, and the floor is reached once $\varepsilon_0^{\text{RESIDUAL}} \gamma^{n_g} \leq \varepsilon_{\min}$.

The aggregate phase is outside the hypothesis: it replaces V by a value lifted from group representatives, which is not an application of T and can raise the span. The envelope is therefore a claim about the stretch in which the rule is still setting the width, and it is verified empirically over exactly that stretch, at every recorded iterate of all three mixes.

B Work-indexed trajectories

Rules reaching $\varepsilon_{\min}$ have similar error floors — inventory $N = 3, Q = 10, \gamma = 0.95$

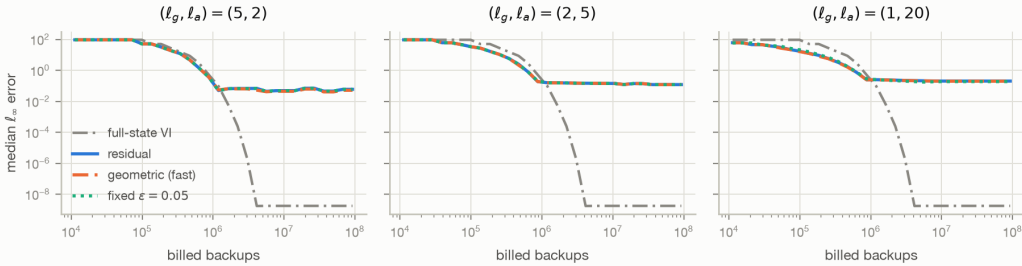


Figure 1: Median value error against billed Bellman backups; terminal estimates are retained past an arm’s 10,000-iteration horizon. Residual, fast geometric, and fixed-0.05 nearly coincide within each mix. The deterministic VI trajectory is repeated for reference; its convergence check stops after 454 sweeps (4.20M backups) though the plotted diagnostic runs to the common horizon.

The flat right-hand tails are held terminal values, not continued computation: each arm stops at its own horizon (67.7M, 30.2M, and 9.2M billed backups at (5, 2), (2, 5), and (1, 20)), and the plotted value is carried forward so the three panels share an axis. Comparisons at budgets past an arm’s own horizon are therefore not work comparisons, which is why the ladder in Appendix D stops at 3×10^6 .

C Inventory instance

The five joint quote-aggressiveness actions have fill probabilities (.1, .2, .4, .6, .8) and spreads (1, .8, .5, .3, .15), so aggressive quotes fill often at a worse price. The one-period cost is

$$c(q, a) = 0.02 q^\top \Sigma q - \delta_a f_a L(q),$$

where δ_a and f_a are the spread and fill probability of action a , Σ has unit diagonal and correlation 0.5 so inventory risk is correlated across the three assets, and $L(q)$ removes fills that would leave the box $\{-10, \dots, 10\}^3$. Costs are rescaled after solving so that $\|V^*\|_\infty = 100$, which puts ε , the equivalence margin, and every reported error on one interpretable scale.

D Budget-indexed differences

Table 2: Median paired difference in value error at fixed compute budgets, as a percentage of the control’s error there, over 20 seeds; negative favors residual. RESIDUAL–FIXED isolates refinement scheduling, RESIDUAL–GEOMETRIC the feedback signal given the schedule. Budgets were fixed before the differences were inspected, but are not preregistered endpoints.

Mix	Contrast	10^5	3×10^5	10^6	3×10^6	final
(5, 2)	RESIDUAL–FIXED	+0.0	+0.2	−0.1	−0.0	+0.0
(5, 2)	RESIDUAL–GEOMETRIC	−0.1	−0.3	+4.1	+6.3	+14.5
(2, 5)	RESIDUAL–FIXED	+0.1	−23.5	−14.1	+0.4	+0.6
(2, 5)	RESIDUAL–GEOMETRIC	+0.1	+0.1	+0.5	+0.0	−0.0
(1, 20)	RESIDUAL–FIXED	−18.8	−5.1	−5.4	+10.2	+6.5
(1, 20)	RESIDUAL–GEOMETRIC	+4.9	+12.2	+3.2	−0.1	−0.1

Each cell is the median over 20 seeds of the per-seed difference, divided by the control’s error for that seed at the same budget; forming the ratio per seed before aggregating preserves the pairing. Errors are read as a step function of billed backups: the error the budget actually delivered, not the one the run was on its way to. The ladder stops at 3×10^6 because the cheapest mix, (1, 20), bills only 9.2M backups in total, and rungs past an arm’s own horizon would compare held terminal values rather than work. Absolute differences and seeded 10,000-resample percentile-bootstrap intervals for both the absolute and relative medians are in the artifact.

E The transplanted slow schedule

The preregistered slow geometric arm used 1,429 aggregate cycles, exactly the number induced by 10,000 iterations at cycle length seven. Transplanted unchanged to (1, 20), whose cycle length is 21, it receives only 477 aggregate entries and is still decaying at the horizon: it ends at $\varepsilon = .232$ and error 1.59, an order of magnitude worse than every other arm. Recalibrating only its cycle count to 477, changing nothing else, ends it at $\varepsilon = .05024$ and error .209, back in the neighborhood of residual and fast geometric. The apparent failure was an unfinished timetable, not missing feedback, and it is the reason the recalibrated arm rather than the frozen one is carried as the slow control at mixes other than the preregistered (2, 5).